Name: Date: Period:

Motion #3

Acceleration Practice

A brand new ride has opened at King's Island. Use the information provided to answer the questions.

1. The ride starts at rest and then gets up to a speed of 50 m/s in 1.9 s. What is its acceleration?

2. Once it is traveling at $50\,\mathrm{m/s}$, it accelerates at a rate of $-9.2\,\mathrm{m/s^2}$ for $3.0\,\mathrm{s}$. What is its velocity after this time?

3. The ride now maintains a constant velocity for 10 s. What is its acceleration during this time?

4. Finally, the ride comes to an abrupt a stop in $0.88\,\mathrm{s}$. What is its acceleration?

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Mixed Motion Practice

Please use the proper problem-solving method. (1) Draw a picture; (2) knowns & unknowns; (3) pick an equation; (4) plug & chug; (5) answer with units.

$$v = \frac{d}{t} \qquad a = \frac{\Delta v}{t} \qquad \Delta v = v_f - v_i$$

5. How long does it take for a car that is traveling 35 m/s to drive **23 km**?

6. What is the final velocity of a roller coaster that starts at rest and accelerates at a rate of $5\,\mathrm{m/s^2}$ for 0.8 s?

7. How far can a person go if they run at a velocity of 8 m/s for **3 minutes**?

8. How much time would it take an object accelerating at $9\,\mathrm{m/s^2}$ to go from 25 m/s to 56 m/s?

9. After accelerating at a rate of $-5\,\mathrm{m/s^2}$ for 8 seconds, you are now traveling at 11.3 m/s. How fast were you going before you started accelerating?